

Project Title : Work with money, not work for money

Team Name : Black Guard

Module Group	Admin No	Full Name
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BUSINESS UNDERSTANDING

Background

We are Black Guard, one of the world's leading banking and investment companies. Over the years, our company has built itself a reputation for assisting clients in securing and growing their wealth. Our company's expertise lies in asset management and we have made ourselves a trusted partner, particularly for older individuals nearing retirement who have amassed a large amount of savings. These clients have benefited from our company's custom investment plans, designed to provide stability and growth as they approach retirement.

However, in the constantly evolving and changing economic landscape, we face a problem in the form of a limited clientele. Since most of our clients stem from an older demographic, they lack the potential for long term investment growth which can only be achieved if our clients started their investment journey earlier. Because of the difference in financial expectations and needs of the younger client base, our project aims to help Black Guard adapt, to attract and retain a younger audience who can make full use of the benefits of early investing.

Business Objective

Our main goal is to attract a younger and more diverse client base by emphasising the benefits of early and long term investing. With this goal in mind, this project also aims to give Black Guard some leverage amongst the investment companies by acting as the go to partner for individuals just starting their careers. In shifting the clientele, we also hope to build long lasting relationships that can contribute to sustained business growth in the coming decades.

Business Success Criteria

In order to measure the success of our initiative, we will take into consideration and monitor the increase of younger customers to Black Guard post launch, reflecting our success in increasing our client base. Additionally, client participation rates will also be tracked through their involvement in early investment programs, which are tailored specifically to appeal to the young investors. Finally, observation of customer retention rates will also be put into consideration though it is important to note that significant improvements will take some time to become evident.

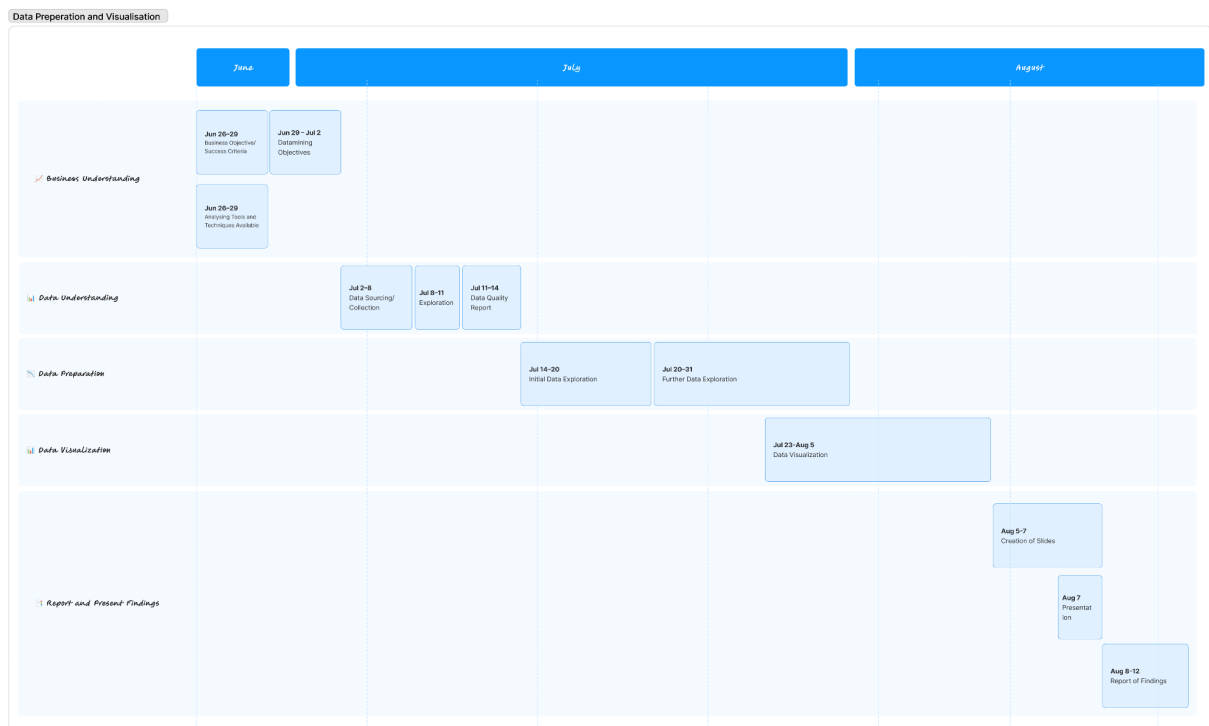
Data Mining Objectives

Goals

This project's data mining objectives are centred on 4 main areas. The first objective is to conduct demographic analysis, finding potential market spaces where we can start advertisement of our initiative. This involves the analysis of income levels, job roles and most importantly geographical locations to make our service globally accessible. The second objective is to analyse investment returns to show the potential for exponential wealth growth over time. This involves the analysis of the trends of the S&P 500 index, visualising the market direction. The analysis will also look deeper into identifying the sectors of the industry which are growing the fastest to provide insights for further analysis. The second goal also makes use of the analysis of market capitalisation to observe and compare the sizes of each company. The third goal makes use of analysing the wealth accumulation of traditional savings along with strategic investments to compare the differences between the 2, further emphasising the benefits of early and long term investing.

The fourth goal involves risk assessment, evaluating market volatility, news sentiment data, and other factors that may contribute to the change of stock prices. Doing so will prove effective in ensuring our clients that we use data-driven insights to guide their investments.

Project Plan



Tools and Techniques

Tools

Power BI - A data visualisation and business analytics tool that enables users to create interactive dashboards and reports from various data sources. As such we used it to create a dashboard so that our clients could interact with the

Google Colab - Google Colab was used as the environment for python development in which we used the various python libraries.

Pandas - We used pandas for basic data storage in data frames, and other simple operations.

Sklearn - Scikit-learn (sklearn) is a Python library for machine learning, providing simple and efficient tools for data mining, data analysis, and model building. Some of the models used are linear regression and cointegration model.

Statsmodel - this is a tool that allows for estimations and the testing of statistical models. Has the capabilities to conduct linear and nonlinear regression, time series analysis, hypothesis testing, and other statistical techniques.

Nltk - Natural Language Toolkit is a python library for processing and analyzing human language. It provides tools for text processing, tokenization, stemming, tagging, parsing, and semantic reasoning, commonly used in natural language processing tasks. This will be used to perform natural language processing on the news articles.

Spacy - spaCy is a fast and efficient Python library for natural language processing, offering tools for tokenization, part-of-speech tagging, named entity recognition, and dependency parsing. It's designed for large-scale text processing and is widely used in NLP applications. Due to the large scale of news articles, this library will be useful to analyse non vital sections of the large news articles.

DATA UNDERSTANDING

Data Collection

Available Data Sources: To conduct our analysis, we sourced data from multiple credible sources, ensuring that our dataset is both viable and up-to-date. The primary sources of our data include sources like Kaggle, Yahoo Finance, BBC, The New York Times, and Reddit, each offering data useful for our objective.

Kaggle:

- Global Salary Data and Salary by Job Title and Country: Both these datasets helped give us insight into the financial ability in terms of different regions and job role personnel, which was then used to base our investment recommendation customised for different demographics.
- BBC & NYT News Articles: The data sets used in this analysis are financial news and global events from BBC and NYT. These sources were important as they would aid in the sentiment analysis and how the news translates to affect the stock prices and general market sentiments. They were drawn from reputable sources, and datasets of this kind make a good platform for realising informed predictions.
- Reddit World News: Financial communities on Reddit can be an extremely rich source of public sentiment related to market moves. Analysing posts and discussions on platforms, such as Reddit, provides insight into the collective mindset of retail investors, often signalling a lead on wider market trends.

Yahoo Finance:

- Stock Data: Yahoo Finance provided necessary data on stock performance, especially for more recent times. This source is constantly updated, ensuring that our analysis remains relevant to current market conditions.

- **S&P 500 Stocks:** The data given is quite detailed in terms of stock performance. The S&P 500 is seen as a general sense of the United States stock market, so it acts as a good indicator for trends in investments. This dataset was chosen on the basis of how detailed it could be over long periods, which would be useful for an analysis of historical trends and the making of future projections. Data from Adjusted Close Price is particularly valuable because it gives the right value of stocks after distribution of dividends, stock split, and other corporate actions, which give a very good sense of performance.

United Nations Development Programme (UNDP):

- **Human Development Index (HDI) Data:** HDI data were sourced from the UNDP and formed a critical part of our demographic analysis. The HDI provides a composite measure of country social and economic development based on life expectancy, education attainment, and income indicators. We've applied this information to finding those nations that have elevated human development, thus indicating that their populations will be more likely to understand and realise the benefits of early investment strategies. The use of HDI allows us to focus those efforts on the regions where the likelihood of financial literacy and investment participation is higher, which makes it a critical component of our overall strategy.

Constraints with Retirement Survey Data: While we had the advantage of getting various datasets, we had a problem in getting retirement survey data and especially current retirement survey data. Most available datasets were either outdated or did not have the granularity we required for our particular demographic analysis. Hence, we focused on financial data and news sentiment, as it is more readily accessible and relevant to the investment based nature of our project.

Why These Data Sources: These are the specific datasets chosen because they cumulatively provide a useful picture of the market trend, financial capacity, and public sentiment, all of which will play a crucial aspect in our analysis. Historical data allows the trend of any particular observation, updated information ensures it is currently relevant, and sentiment data lets one predict future movement, all combined to form a wholesome basis for our investment recommendations. The use of these sources solidifies our analysis with credible, diverse, and pertinent information; it helps to increase the credibility and truth of what is found.

Data Dictionaries

Habib - Industry_market_cap_with_others

Columns	Datatype	Description
---------	----------	-------------

Industry	Object	Used to identify which industry the company belongs to.
Company	Object	Ticker of each company name to help identify which company it is.
Market Cap(Millions)	float64	Market capitalization value in millions
Slope(2008-2024)	float64	The normalised rate of growth of each company from 2008 to 2024
Entity	Object	The company name of the entity. It is linked to the company column.

Habib - Industry_slope_average

Columns	Datatype	Description
Industry	Object	Used to identify which industry the slope belongs to.
Slope(2008-2024)	float64	The normalised rate of growth of each industry from 2008 to 2024

Habib - stocks_date

Columns	Datatype	Description
Adj Close	float64	Value in dollars of adjusted close of the company for a particular date.
Company	object	Ticker of each company name to help identify which company it is.name to help identify which company it is.

date	object	The date of the entry
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Alex - investable_amount_by_country_top50

Columns	Data types	Description
Country	object	Country that can invest
Cost of living	float64	The cost of living in that country
median_salary	float64	The average salary of someone living in that country
Remaining amount	float64	The disposable income
Investable amount	float64	The amount they can invest

Alex - top_3_jobs_by_salary_per_country

Columns	Data types	Description
Country	object	The country that job statistic is from
Job Title	object	The name of the job
Salary	float64	The salary of that job

Alex - top_50_countries_gnipc/hdi/le/mys

Columns	Data types	Description
Country	object	The country for the specific data entry
gnipc/hdi/le/mys	float64	Gross national income / human development index / life expectancy / mean year of schooling
gnipc/hdi/le/mys_year	int64	The year of entry for that particular data value

Min - news.csv

Columns	Data types	Description
---------	------------	-------------

Date	Date	yyyy-mm-dd
neg	float64	Negative measure of news article
neu	float64	Negative measure of news article
pos	float64	Negative measure of news article
compound	float64	Negative measure of news article
processed_news	object	Keywords within the week (regex removed)
sentiment	object	POSITIVE, NEGATIVE
news	object	A description or abstract of the news article

Min - stocks_trends

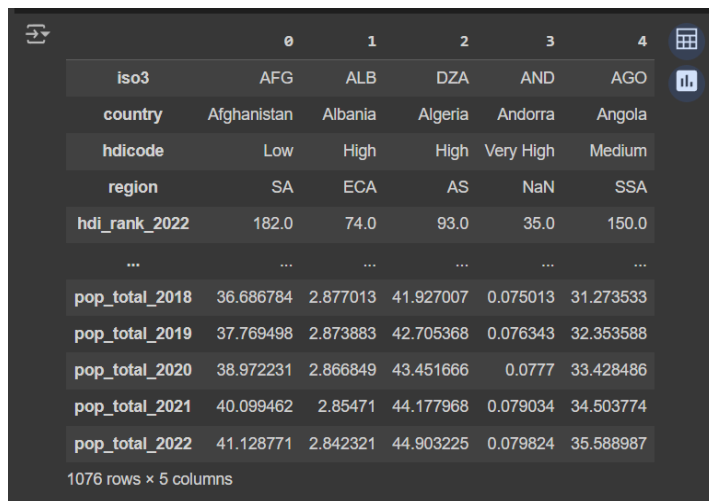
Columns	Data types	Description
Date	Date	yyyy-mm-dd
Ticker	Object	Symbol of the company
Overall PCT Change	float64	Percent change of adj close price based off its initial value at 100%

Min - metrics

Column	Data Type	Description
Ticker	object	Symbol of the company
Annualised Return	object	Annual Return of the company over the whole period
Annualised Volatility	object	Annual Volatility of the company
VaR	object	Value at Risk of the investment at 95% confidence interval

Exploration

HDI from UNDP:

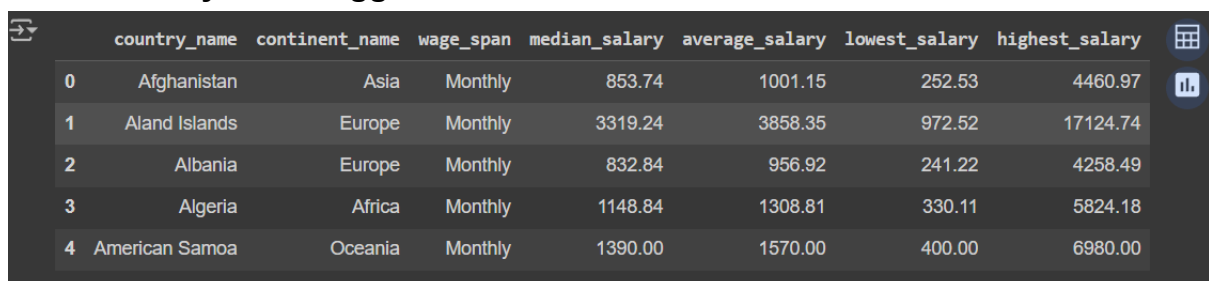


	0	1	2	3	4
iso3	AFG	ALB	DZA	AND	AGO
country	Afghanistan	Albania	Algeria	Andorra	Angola
hdi	Low	High	High	Very High	Medium
region	SA	ECA	AS	NaN	SSA
hdi_rank_2022	182.0	74.0	93.0	35.0	150.0
...	...	...	...	...	...
pop_total_2018	36.686784	2.877013	41.927007	0.075013	31.273533
pop_total_2019	37.769498	2.873883	42.705368	0.076343	32.353588
pop_total_2020	38.972231	2.866849	43.451666	0.0777	33.428486
pop_total_2021	40.099462	2.85471	44.177968	0.079034	34.503774
pop_total_2022	41.128771	2.842321	44.903225	0.079824	35.588987

1076 rows × 5 columns

Large number of columns in the data cause difficulties in understanding which columns are useful for this context. Dataframe in the picture above is transposed and thus the shape of the data frame is (5,1076).

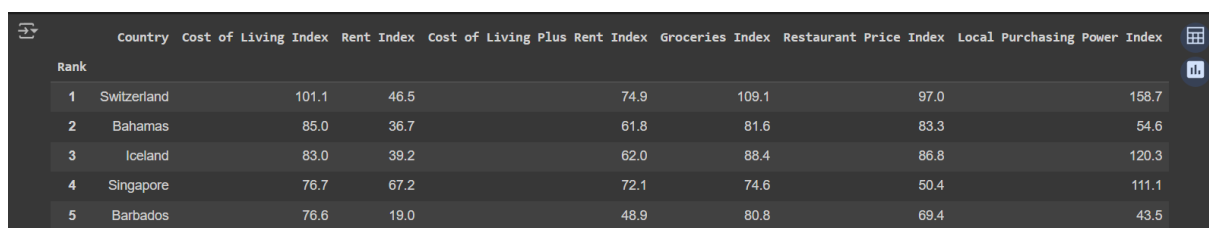
Median Salary from Kaggle



	country_name	continent_name	wage_span	median_salary	average_salary	lowest_salary	highest_salary
0	Afghanistan	Asia	Monthly	853.74	1001.15	252.53	4460.97
1	Aland Islands	Europe	Monthly	3319.24	3858.35	972.52	17124.74
2	Albania	Europe	Monthly	832.84	956.92	241.22	4258.49
3	Algeria	Africa	Monthly	1148.84	1308.81	330.11	5824.18
4	American Samoa	Oceania	Monthly	1390.00	1570.00	400.00	6980.00

Lots of unnecessary columns such as continent_name as well as the other salary metrics aside from media_salary which were not required for the given context.

Cost of Living from Kaggle



	Country	Cost of Living Index	Rent Index	Cost of Living Plus Rent Index	Groceries Index	Restaurant Price Index	Local Purchasing Power Index
Rank							
1	Switzerland	101.1	46.5	74.9	109.1	97.0	158.7
2	Bahamas	85.0	36.7	61.8	81.6	83.3	54.6
3	Iceland	83.0	39.2	62.0	88.4	86.8	120.3
4	Singapore	76.7	67.2	72.1	74.6	50.4	111.1
5	Barbados	76.6	19.0	48.9	80.8	69.4	43.5

Lots of unnecessary columns as only the Cost of living index column as well as the country column were required. The rest of the columns are not required for the given context.

Income and Job Roles from Kaggle

SP500 stocks from yfinance

```
1 #find missing companies that were unable to be downloaded
2 missing_companies = set(com_list['Symbol']) - set(stocks_data['Company'].unique())
3
4 missing_companies
```

{'BF.B', 'BRK.B'}

Dataset is missing data for entities Brown-Forman Corporation Class B and Berkshire Hathaway Inc Class B.

	Open	High	Low	Close	Adj Close	Volume	Company
Date							
2008-01-02	70.434784	70.869568	68.754181	69.155518	41.474415	5326625.0	MMM
2008-01-03	69.247490	69.799332	68.871239	69.147156	41.469398	3258024.0	MMM
2008-01-04	68.561874	69.113708	68.185616	68.361206	40.998043	4329879.0	MMM
2008-01-07	68.645485	68.687294	67.533447	67.817726	40.672112	5051665.0	MMM
2008-01-08	68.018394	68.620399	66.973244	67.065216	40.220810	6830595.0	MMM

Lots of unnecessary columns such as the adj_close column as well as the tickers are required. The rest of the columns are not required for the given context.

SP500 company profile from Wikipedia

	Symbol	Security	GICS Sector	GICS Sub-Industry	Headquarters Location	Date added	CIK	Founded
0	MMM	3M	Industrials	Industrial Conglomerates	Saint Paul, Minnesota	1957-03-04	66740	1902
1	AOS	A. O. Smith	Industrials	Building Products	Milwaukee, Wisconsin	2017-07-26	91142	1916
2	ABT	Abbott	Health Care	Health Care Equipment	North Chicago, Illinois	1957-03-04	1800	1888
3	ABBV	AbbVie	Health Care	Biotechnology	North Chicago, Illinois	2012-12-31	1551152	2013 (1888)
4	ACN	Accenture	Information Technology	IT Consulting & Other Services	Dublin, Ireland	2011-07-06	1467373	1989
...	...	...	...	...	...	...	...	...
498	XYL	Xylem Inc.	Industrials	Industrial Machinery & Supplies & Components	White Plains, New York	2011-11-01	1524472	2011
499	YUM	Yum! Brands	Consumer Discretionary	Restaurants	Louisville, Kentucky	1997-10-06	1041061	1997
500	ZBRA	Zebra Technologies	Information Technology	Electronic Equipment & Instruments	Lincolnshire, Illinois	2019-12-23	877212	1969
501	ZBH	Zimmer Biomet	Health Care	Health Care Equipment	Warsaw, Indiana	2001-08-07	1136869	1927
502	ZTS	Zoetis	Health Care	Pharmaceuticals	Parsippany, New Jersey	2013-06-21	1555280	1952

503 rows x 8 columns

Lots of unnecessary columns which were not required for the given context

Reddit World News from Kaggle

No issues were found with the dataset

BBC News from Kaggle

The files were too big and the headers were too long causing issues.

NYT News from Kaggle

The files were too big and the headers were too long causing issues.

Data Quality

HDI from UNDP:

This dataset had all the correct data types, and was formatted correctly. Data was slightly incomplete as the UNDP had not collected data from certain countries. The data was consistent and showed no signs of wild fluctuations without reason. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

Median Salary from Kaggle:

This dataset had all the correct data types, and was formatted correctly. Data was complete. As data is taken from one year, there is no need to check for consistency. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

Cost of Living from Kaggle:

This dataset had all the correct data types, and was formatted correctly. Data was complete. As data is taken from one year, there is no need to check for consistency. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

Income and Job Roles from Kaggle:

This dataset had all the correct data types, and was formatted correctly. Data was complete. As data is taken from one year, there is no need to check for consistency. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

SP500 stocks from yfinance:

This dataset had all the correct data types, and was formatted correctly. Data was slightly incomplete as there are 2 companies with missing data. The data was consistent and showed no signs of wild fluctuations without reason. Some . Data was consistently formatted for all rows.

SP500 company profile from Wikipedia:

This dataset had all the correct data types, and was formatted correctly. Data was complete and had no missing entries. The data was consistent and showed no signs of wild fluctuations without reason. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

Reddit World News from Kaggle

This dataset had all the correct data types, and was formatted correctly. Data was complete and had no missing entries. As each entry is a news article, . Each record was unique and was not duplicated. Data was consistently formatted for all rows.

BBC News from Kaggle

This dataset had all the correct data types, and was formatted correctly. Data was complete and had no missing entries. The data was consistent and showed no signs of wild fluctuations without reason. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

NYT News from Kaggle

This dataset had all the correct data types, and was formatted correctly. Data was complete and had no missing entries. The data was consistent and showed no signs of wild fluctuations without reason. Each record was unique and was not duplicated. Data was consistently formatted for all rows.

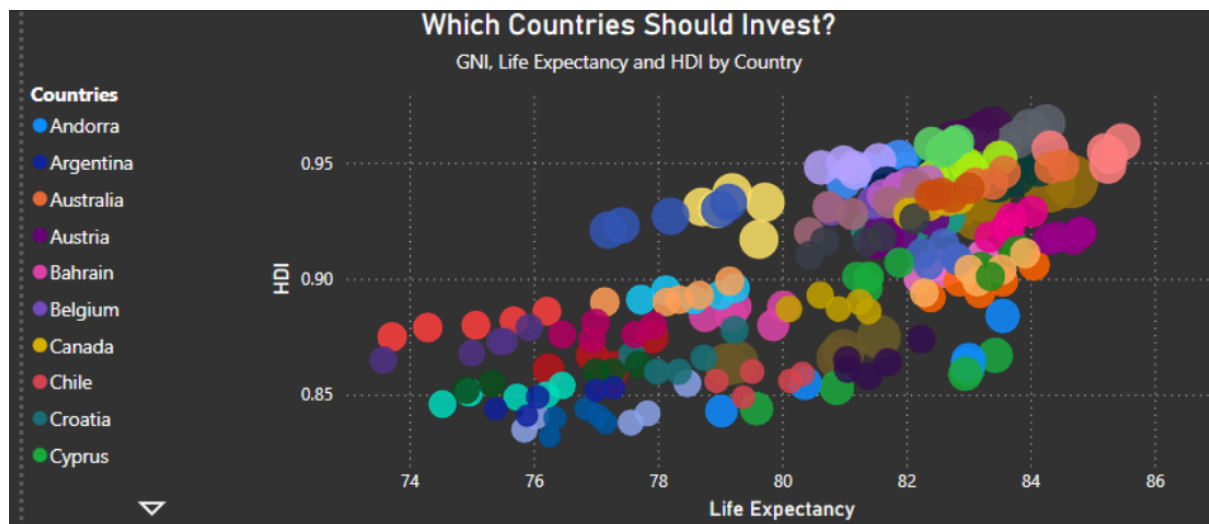
Data Visualization

Dashboard #1 (Alex): Describe the general demographics of working people for potential customer segmentation.

Objective

The primary objective of this dashboard was to find potential markets and target demographics to widen our client base, mainly focusing on countries which were considered more developed based on the HDI index along with which job roles in these countries provided appropriate salaries with disposable income necessary for investing.

Chart 1: Compare and contrast the demographics and financial awareness of the working population across the globe.

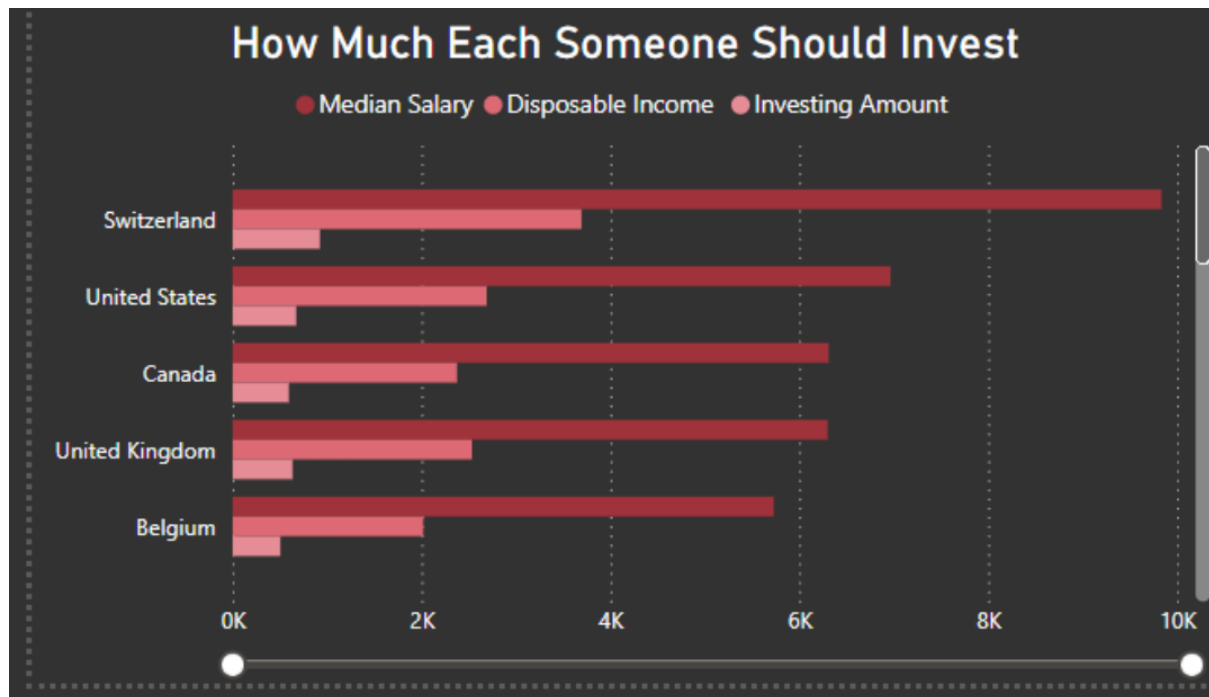


Data Used: This chart utilises data on Gross National Income (GNI) per capita and Life Expectancy by country, The Human Development Index (HDI) sourced from the United Nations Development Programme (UNDP). HDI is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable and having a decent standard of living. The HDI is the geometric mean of normalised indices for each of the three dimensions. We considered HDI to be suitable for our analysis as it provides a clear picture of a country's socio-economic development. It is generally assumed that citizens originating from countries with high HDI have both better and easier access to education, higher incomes and higher life expectancy. Considering these factors we deduced that these people would be more likely to understand and appreciate the benefits of early investment. As such, we filtered out the top 50 countries by HDI, life expectancy and gross national income to consider as our foundation for advertising our service. Among the top 50 are some nations like Norway, Switzerland, Ireland, Germany and even Singapore, all of which hold consistently high HDI scores.

Purpose of the Graph: A scatter plot is used here to visualise the relationship between life expectancy, GNI per capita, and HDI across different countries. With HDI across the x-axis, life expectancy along the y-axis and the GNI representing the size of each point plotted. By mapping these variables together, the graph highlights which countries have populations that are not only financially suitable but also

socially and economically well off to take advantage of our service. The plot allows for easy identification of countries where our targeted investment campaigns may have the most impact, making it a useful tool for strategic decision-making.

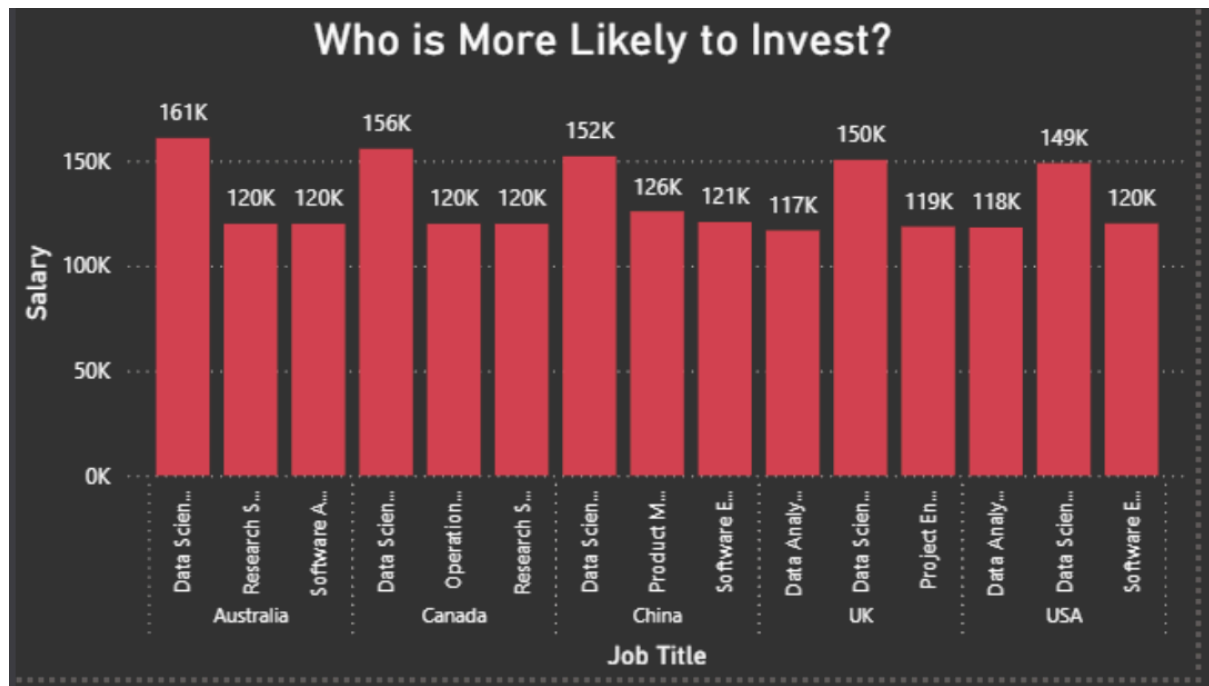
Investment Amount Calculation



Data Used: To determine an appropriate investment amount for individuals in various countries, we utilised global salary data with the Cost of Living Index (CoLI). The CoLI dataset provided information where a CoLI of 100 equated to \$1,640.90 USD. By relating these two datasets, we were able to subtract the cost of living for each country from its median salary, allowing us to make informed assumptions about an individual's disposable income in each region and upon further research and discussion, we decided on 25% of that disposable income amount to be deemed as investable income.

Purpose of the Graph: A bar chart is used due to its ability to illustrate the relationship between median salary, disposable income, and the recommended investment amount across different countries. With median salary and disposable income plotted along the x-axis, and the investing amount shown as a portion of disposable income, the chart provides a visual of how much someone in these countries could realistically invest.

Job Role Analysis



Data Used: To find out who is more likely to invest within a given country, we utilised global job salary data. This dataset provided information on young professionals just entering the job market, including their respective salaries, occupations, years of experience, and country of origin. We specifically filtered the data to focus on those with 10 years of experience or less, grouping them by country and arranging the results by salary. This allowed us to identify the job roles that offer higher salaries within each country, making these professionals candidates for early investment opportunities.

Purpose of the Graph: The bar chart visualises the relationship between job roles and salary across different countries, highlighting which positions are most attractive to young professionals. By targeting individuals in these high-paying roles with tailored investment opportunities, we significantly increase the likelihood of attracting new clients. The chart serves as a tool to identify and focus on the most promising jobs within each country.

Conclusion: The data from the three graphs clearly shows the demographic, financial, and professional profiles of those who could benefit from early investment strategies. First, the Demographic Analysis using HDI, GNI per capita, and life expectancy identified financially capable and socio-economically advanced countries for targeted investment campaigns, focusing on high-HDI regions where individuals are more likely to appreciate early investing.

Second, the Investment Amount Calculation emphasised considering both salary and cost of living to determine realistic investment amounts, making our advice practical and actionable.

Finally, the Job Role Analysis highlighted which professions in various countries have the financial means to invest, particularly young professionals in high-income roles, increasing the chances of attracting a younger client base to Black Guard.

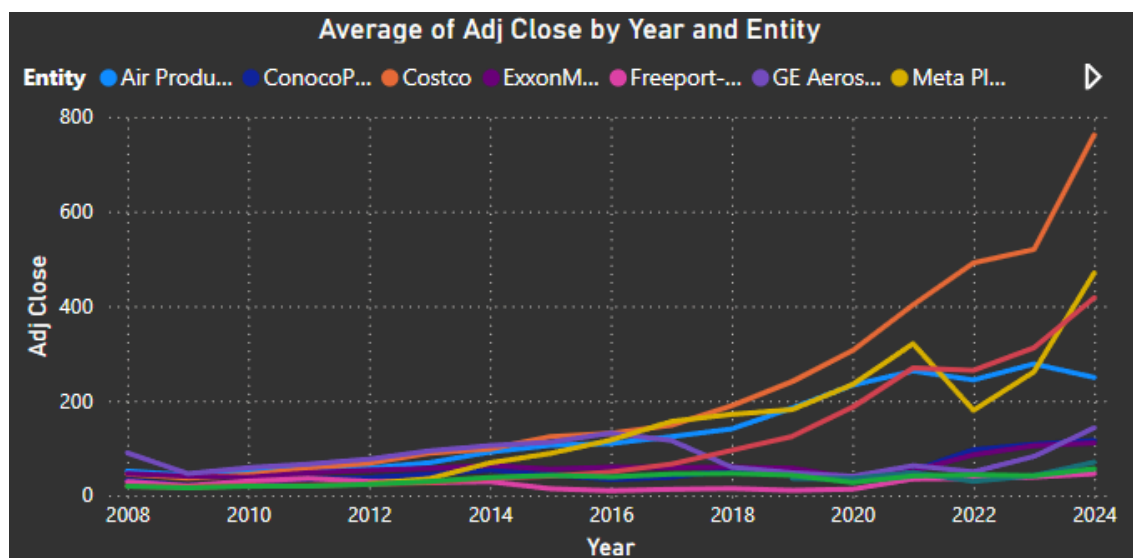
Overall, these insights confirm that we've successfully identified the demographic most likely to benefit from early investing, positioning Black Guard to expand its client base and attract younger investors by integrating these findings into our marketing and client engagement strategies.

Dashboard #2 (Habib): Analyse investment returns to show the potential for exponential wealth growth over time.

Objective

The primary purpose of this dashboard is help clients visualise the various different aspects of S&P500. These aspects include, general growth trend of companies, comparative market size and lastly the growth rate of each industry component in S&P500. This in turn will provide insight into when clients should invest into the index as well to showcase the potential growth early investment can provide.

Chart 1: Analysis of the overall trends of S&P500 to visualise the general trend of the index



Data Used/Purpose of Graph

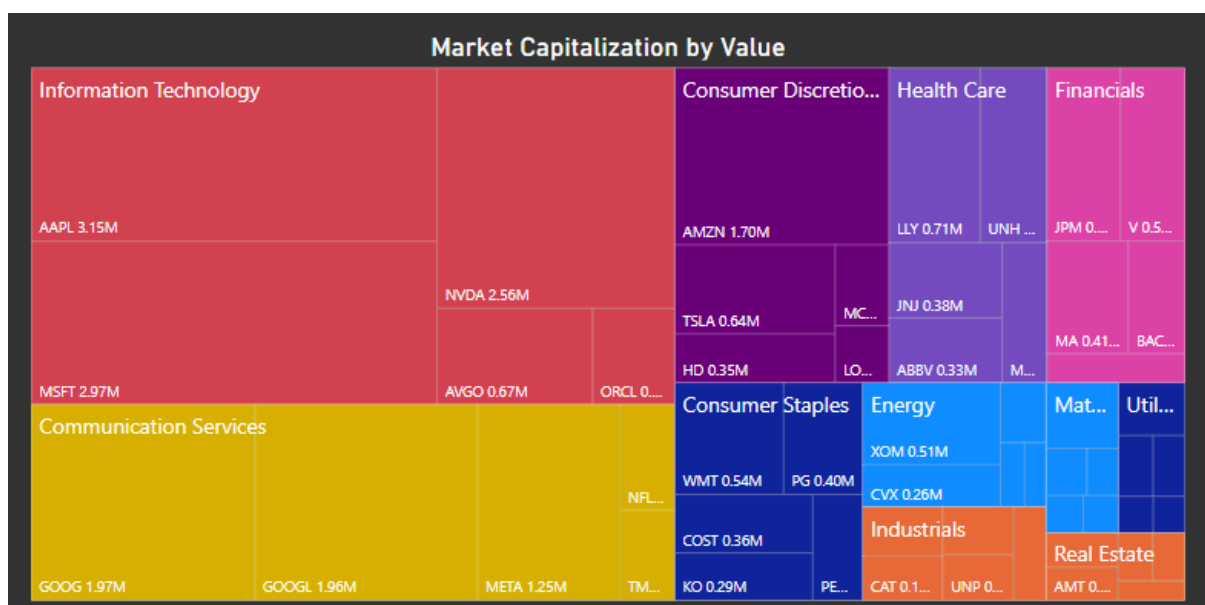
One of the main indicators of value of a stock is the adjusted close value. Adjusted close is the final price of a stock on any given day, it is calculated after closing price and hence is final. It is calculated after taking into account all external factors such as stock splits, dividend payouts and other changes to the stocks. As such, it is the best indicator to use for historic examination as it gives an accurate representation of a firm's equity value. As such, we used the adjusted close value over time to showcase the growth of entities in the S&P500. This in turn will showcase to our

clients who are young working professionals that historic data proves that the index will exponentially grow over a long period of time. To facilitate interests client's may have, this graph is able to be sliced via industry and or by company, to allow clients to be able to view individual/groups of companies separately if they wish to invest in said companies separately. There is also a filter applied on the entire graph to sort by growth rate so as to show the best performing companies in terms of stock value growth.

Insights

From the graph, we can note there is a significant increase in almost all companies adjusted close value. Some companies to note are Costco in orange, META platforms in yellow and Microsoft in red. These are companies who have shown considerable growth and have a significant market share in their respective industries. Clients will also be able to interpret the various different global events that caused stock fluctuations, most recent being Covid-19 which can be seen for some companies around the 2019-2022 companies. This can help clients understand the index better and be able to know with moderate confidence which companies are less prone to stock value crash.

Chart 2: Visualisation of the market capitalization of each company



Data Used/Purpose of Graph

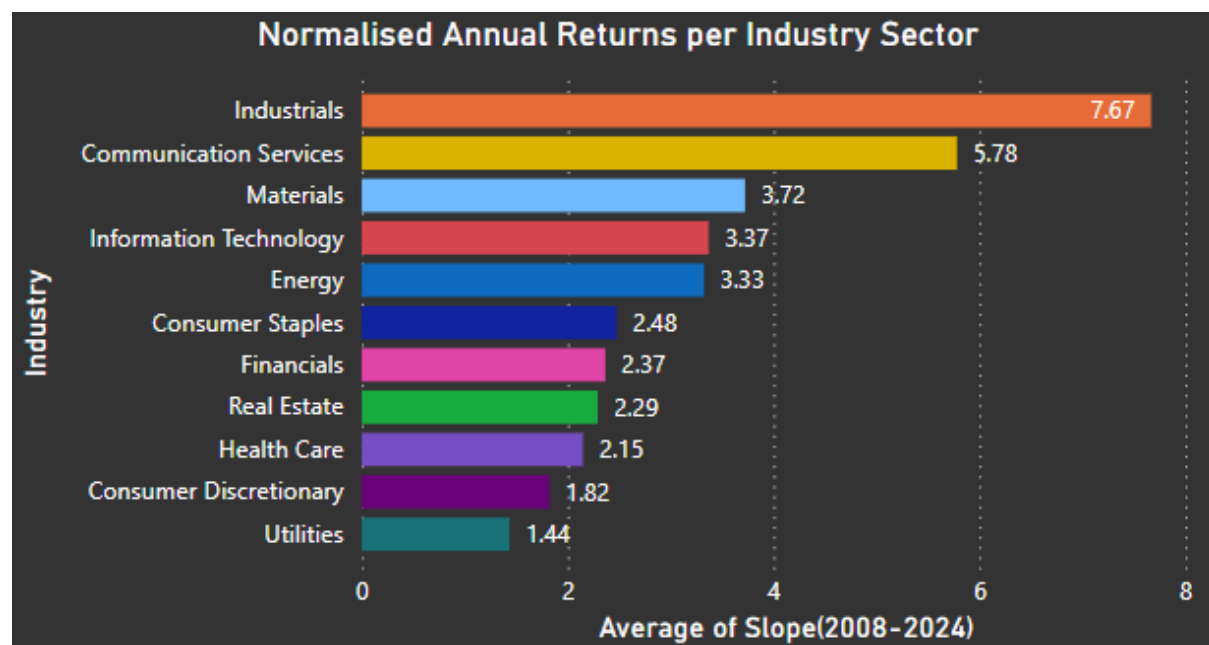
One of the main indicators of an entity's presence in the market is the market capitalization value. It is derived by multiplying outstanding shares; the shares in the possession of stakeholders, by the value of one share. What this does, is help our

clients to be able to understand approximately how entities in the S&P500 index would affect the index. A larger company such as Apple would cause a much larger shift in the index's value as compared to Wells Fargo even if the rate of growth is lower. This is due to the fact that comparatively, the sizes of the companies are significantly different.

Insights

From the graph, we can see that almost all of the industries are dominated by one or two companies. As such, clients will be able to interpret that there are a few key companies to look out for to understand the overall value of the S&P500 index which they can take note off via this graph. Some notable companies with the highest market capitalization values are Apple(APPL), Microsoft(MSFT), Nvidia(NVDA) which is also an extremely fast growing company due to advances in AI technology they have made. A notable Company to mention is Google as they have two listed entities as seen in the communication services in yellow. Effectively making them the largest overall company albeit split up in two entities.

Chart 3: Analysis of individual industry sectors



Data Used/Purpose of Graph

The purpose of this graph is two fold, one is to help clients visualise the current fastest growing industries. Secondly, it is to help average out the various different growth rates of each company to assist clients in seeing the overall trend of each industry in S&P500 and as an extension, the general trend of S&P500. This graph is to help reinforce the sentiment that early investment will lead to growth of our clients.

Insights

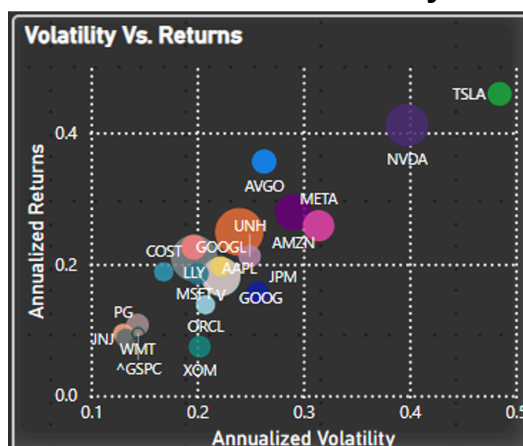
From the graph, we can infer that over the time frame of 2008-2024, the most grown industry is the industrial sector. We can also see that the majority of the sectors have grown by around 2.15 - 3.70 units. The only large sector with a large market capitalization as well as high growth is the Communication Services sector. This likely has to do with the increased usage of the internet compared to the early 2000s. As such, if the client wants to exponentially grow their value, they can selectively invest in specific industries if they wish to pick and choose what stocks they wish to invest in. If not, this shows at the bare minimum to ensure they have a return on their investment, they can invest in the S&P500 index and make turn a profit.

Dashboard #4: (Min <3)

Data Mining Goal: Identify correlations between investment returns and news breakout to determine the best and worst scenarios for public risk awareness

Objective: This dashboard is to highlight how volatile the S&P500 is in regards to news breakout so that the potential clients can make informed-decisions for themselves.

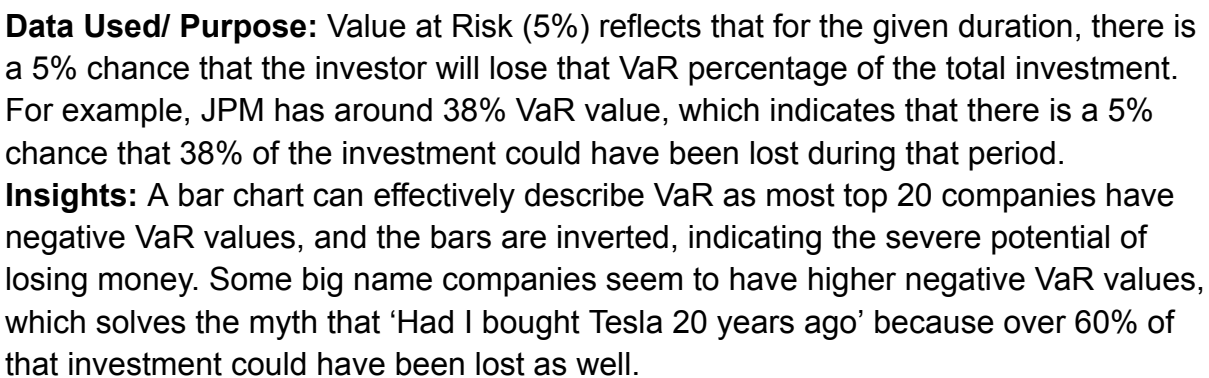
Chart 1: Annualized Volatility vs. Returns



Data Used/ Purpose: A scatter plot with varying bubble sizes has been used in this chart. The x-axis represents the annualized volatility of top 20 S&P500 companies while the y-axis is for annualized returns. The size of each dots reflects the market capitalization of each company.

Insights: Using a scatter plot was a good choice as the individual companies with their annualized returns and volatility can be clearly seen. For example, TSLA which is at the top right corner has a high returns but at the same time has the highest volatility, while those clustered at the lower left corner has low returns and low volatility. The linear relations ($y=x$) also highlights that the more returns there are, the more volatile the market is.

Chart 2: Value at Risk (5%)



Overall Trends of Top S&P500 Companies

The chart displays the Average of Overall PCT Change (Y-axis, 0 to 12,000) over time (X-axis, 2010 to 2020). The data series include AMZN, AAPL, LLY, COST, GOOGL, and ^GSPC. All series show a significant upward trend, particularly after 2015, with AMZN and AAPL reaching the highest values by 2020.

News Sentiment Analysis

sentiment	news
NEGATIVE	17 Hit or Stabbed, 7 Confirmed Dead in Tokyo's Akihabara
NEGATIVE	3,000 dead from cholera in Zimbabwe, Local Government takes a
NEGATIVE	conceal the scale of this deadly epidemic'
NEGATIVE	A 'terrorist' being questioned'
NEGATIVE	A Russian scientist this week said Russia has fallen behind the U.S. race; the race to arm sea mammals, that is. It turns out, the U.S. r
POSITIVE	A treasure trove of Tibetan art and manuscripts uncovered in 'sk

Hot Words

The word cloud contains terms such as: two, york, trump, aid, help, one, president, police, world, help, home, nigh, trouble, and others.

Data Used/Purpose: This chart has several components which are all related. The main is the line chart which shows the trends of S&P500 based on its initial value in 2008 at 100%. These trends are affected by news articles, i.e., global events in that period (1 week interval). Selecting a certain time frame will display the news released within that period and the trends will change accordingly. The word cloud also summarizes the news articles by taking out the hottest words within that period. The heatmap beside the news table is to show the breakdown of positive, neural and negative (in that order from top to bottom) of the selected news article.

Insights: As these charts show the relations between text and numerical trends, we could not derive a figure for the correlation, but our several trials showed that the S&P500 stocks are noticeably affected by news breakout, especially with politic news such as those with keywords 'US' and major global events such as covid19 and 2008 recession.

Conclusion: The dashboard successfully depicts the essence of the data mining goal; it informs the user of the risks and the impact of news. The interaction with slicers also help the user explore freely and make informed decisions on his own.

Discussion

During the entire project, we had many challenges that tested the very basis of the application of business concepts and statistical analyses. The biggest difficulty was the understanding of difficult terminologies used in the business world, which was not easy to interpret while analysing financials and investment strategies. There are many terms, like "market capitalization," "adjusted close price," and "cost of living index," that pushed us to look further into the details for our assurance that we were using them correctly in our study and presentation.

Additionally, working with hard statistics proved challenging: deriving meaningful information from large data sets, calculating sound financial projections, and interpreting the exact results required good statistical methods and tools. For some of us, this was a steep learning curve as we moved beyond basic statistics into much more advanced areas including regression analysis, correlation, and data visualisation.

The project was really challenging in this respect, but it really broadened our scope and geared us toward better competence both in business and statistical analysis. We became so much more comfortable with the terminology and with the statistical tools that we were using that we could clearly apply them effectively toward meaningful insights by the end of the project. This aspect really accentuated how important it is to keep on learning and being adaptive, especially in matters concerning complex and technical topics.

Conclusion

Our analysis strictly adhered to the CRISP-DM methodology, making sure that each phase of the project, from business understanding, data preparation, modelling, evaluation, and deployment, was systematically addressed. This guided approach enabled us to derive meaningful insights at each of the points, essential to realising the objectives of our project. The usage of demographic statistics like HDI and GNI helped segment the population with high investment potential, and the analysis of disposable income had placed practical and viable recommendations for

investments. Also, it is the analysis of job roles which makes the professional sectors dominated that the early investment strategies can be most effective. The findings were analysed and interpreted taking a more actionable directive. For instance, the S&P 500 trends and the adjusted close price computed showed very well the opportunities for long-term growth. These indicators are important in developing tailor-made investment models that young professionals can identify with. In summation, this project was executed to a satisfactory level by following the CRISP-DM methodology in order to realise the objectives of giving Black Guard the strategic direction for extending its customer base. The conclusions from our analysis were firmly data-based, underpinned by robust results that are useful in informed decision-making. This attests well to the ability of our team and the application of methodology in the facing of real-world challenges.

References

[Fewer Singaporeans able to spend beyond basics and more are short on emergency funds: OCBC survey](#)

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